

EECE 430 — Final Exam Answer Key

Quick Reference:

Q	Ans	Q	Ans	Q	Ans
1.	B	2.	B	3.	C
4.	C	5.	C	6.	B
7.	C	8.	B	9.	C
10.	C	11.	B	12.	C
13.	C	14.	C	15.	B
16.	B	17.	C	18.	B
19.	B	20.	C	21.	B
22.	B	23.	B	24.	C
25.	B	26.	B	27.	B
28.	B	29.	B	30.	C
31.	B	32.	B	33.	B

Detailed Explanations:

- 1. B** — Refactoring makes future changes easier since changes cannot be reliably anticipated.
- 2. B** — In TDD, after writing a test you run it expecting failure since the functionality isn't implemented yet.
- 3. C** — A big disagreement (2 vs 13) means the team should discuss further, as there are different understandings of the story.
- 4. C** — Evidence suggests a pair may be MORE efficient, not less. It is not always less efficient.
- 5. C** — TDD limitations include difficulty writing incremental tests and judging test completeness.
- 6. B** — Daily scrums are for sharing info, reviewing progress, and identifying problems for short-term replanning.
- 7. C** — The team should select ~21 points based on priority and velocity. Sprints are not extended.
- 8. B** — XP requires all tests must pass for every build; the build is only accepted if tests run successfully.
- 9. C** — Ready for consideration: high-level ideas that may radically change or be excluded.
- 10. C** — Scrum does not eliminate the product backlog — the backlog is central to Scrum.
- 11. B** — Maintainability needs fine-grained components; but inter-component communication hurts performance.
- 12. C** — The application server handles application-specific logic and storage/retrieval requests.
- 13. C** — Repository pattern: tools share data through a central repository (e.g., an IDE).
- 14. C** — Pipe and filter is NOT suitable for interactive systems.
- 15. B** — Stateless means no internal state, enabling replication and migration between servers.
- 16. B** — Hard to predict how many extra columns are needed, and customers may need different data types.
- 17. C** — Multi-tenant: easier update management — update one instance and all customers get it.
- 18. B** — Containers excel at rapid scaling with minimal overhead for independent, lightweight services.
- 19. B** — Scalability handles increasing load; elasticity also includes scaling down when demand decreases.
- 20. C** — Programming language choice is not a SaaS design issue discussed in the course.

- 21. **B** — Integrate and test locally first, then push to the shared repository to trigger CI.
- 22. **B** — Delivery = tested in production replica and ready. Deployment = automatically released to users.
- 23. **B** — Newer source timestamp means changes were made, so the build system recompiles.
- 24. **C** — Recovery: versioned infrastructure models in code management allow rollback to working versions.
- 25. **B** — Healthy DevOps: fewer failed deployments, faster recovery, more frequent deployments.
- 26. **B** — Handling unrelated responsibilities (low cohesion) AND accessing other services' databases (high coupling).
- 27. **B** — Eventual consistency uses a pending updates log that replicas read, apply, and confirm.
- 28. **B** — Circuit breakers immediately block calls to failed services, avoiding per-call timeout delays.
- 29. **B** — Choreography is harder to debug because no central point tracks the overall workflow state.
- 30. **C** — Using a stolen session cookie to impersonate a user is session hijacking.
- 31. **B** — Attackers can deliberately trigger lockouts to deny access to legitimate users.
- 32. **B** — Data at rest should be encrypted; data in use encryption slows performance and prevents general search.
- 33. **B** — Storing data in a country with weaker laws without an agreement violates data protection principles.

— END OF ANSWER KEY —